Smart Home Automation System Using IOT

ABSTRACT

In the blessing field, utilization of home automation is expanding step by step exponentially. Home automation is relate propelling field in information innovation. Home automation is simply dealing with various exercises of home with the new cutting edge innovations. amid this undertaking we tend to are building up an encapsulation to exhibit various parts of Home Automation, similar to overwhelming electronic gadgets, exploitation just whenever required in accordance with the outside conditions, swing on/off the electrical machines with cell gadgets from a different area. It moreover goes for sparing the surroundings through conservative water utilize, hearth recognition, and so on. It allows the proprietor to hold out entirely unexpected exercises of home from a remote place. Home security might be a field of home automation. It focuses on the wellbeing parts of homes and workplaces. this will be accomplished by local systems administration or by remote. These frameworks protect homes from interlopers and thieves. Home security starts with home wellbeing. Home security begins with householders finishing ventures to protect their home and in this manner the those that rest in it. the acknowledgment of home security robotization has been swelled in enormously as of late because of a great deal of higher moderateness and simplicity. Thus, our venture.

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Chapter-1

1.1 Introduction

Home monitoring system and control system are a device that is implemented using Internet of Things. The Internet of Things is the inter-networking of physical devices, vehicles, buildings, and other items embedded with electronics, software, sensors, actuators, and network connectivity that enable these objects to collect and exchange data. The home monitoring system monitors doors and windows of your home and notifies you of any new access of your property via a data feed. Home monitoring system consists of sensors to detect intrusion and captures and sends multiple pictures of the intruder to the user anywhere through Internet.

By the rapid developments of new technologies, monitoring, controlling services have been started to be served along with internet as an instrument providing interaction with machinery and devices. The system can be use in several places like banks, hospital, labs and other sophisticated automated system, which dramatically reduced the hazards of unauthorized entry. The main reason to develop this system is to save time and man power along with maintaining security and convenience.

1.2 Existing System And Proposed System

EXISTING SYSTEM

Design and Realization of Home Appliances Control System Based on The Android Smartphone present the information about the remote appliances control system based on the Android smart phone is designed and realized. A user logs into the smart phone interface, and clicks the buttons gently to send message commands which will be transmitted to home information Center through the GSM network. Then the PIC processor recognizes the specified command, and controls the home appliance switches in the wireless radio frequency manner to achieve remote control of appliances ultimately. The existing system of our project didn't use Internet of things concept to control the home appliances. This is suitable for modern technologies to control devices wherever you located in the world. So, we are moving to the IOT technology as proposed system in our project for the home automation.

DISDAVANTAGES OF EXISTING SYSTEM:

• No remote-control operation
• Depend on others to operate
• No muscle contraction sensing
• Direct pc interaction
• There are various existing technologies available for similar purposes but their cost and complexity is major disadvantage.
• To use system that takes the input from the voice recognition module and uses the microcontroller's intelligence to operate different devices.

PROPOSED SYSTEM

The proposed system designed for security purpose and to monitor the basically consists of Arduino module and sensor. Node MCU is connected to the internet through IOT Technology with sensors to monitor the home efficiently. We connected PIR, Vibration, Gas and LDR Sensors to the Node MCU to monitor our home through IOT Technology. We can control home electrical devices such as bulb, fan, motor through the Internet of Things (IOT) Home monitoring system consists of sensors to detect intrusion and captures and sends multiple pictures of the intruder to the user anywhere through Internet. The system is implemented via the use of IOT, which is the interconnection of machines via Internet for advanced connectivity

ADVANTAGES OF PROPOSED SYSTEM:

• Long distance coverage
• Real time controlling and monitoring
• Control through mobile, pc and also the phablets
• Low cost and expandable allowing a variety of devices to be controlled
• Saves money and energy
• All in one user friendly system
• This is noise free system.
• High secured

Chapter-2: Theoretical Backgrounds

2.1 Introduction

A smart home incorporates sensors, actuators, middleware, and a network and has two major interacting components which is a smart network and a smart load. The Smart home known as House automation, with the use of new technology, to make the domestic activities more convenient, comfortable, secure, and economical.

The system is implemented via the use of IOT, which is the interconnection of machines via Internet for advanced connectivity. This paper presents a low cost and flexible le home control and monitoring system using an embedded micro-web server, with IP connectivity for accessing and controlling devices and appliances remotely. The proposed system does not require a dedicated server PC with respect to similar systems and offers a novel communication protocol to monitor and control the home environment with more than just the switching functionality Home automation is the control of any or all electrical devices in our home or office.

2.2 Background

The process of improving and upgrading the living standard of the house has been raised due to the advanced technology applied in this era society. Home Automation System is implemented for decades but due to the costing and budgeting of the project, it still remains as a niche product for high end consumers. Although the concept of smart home automation has been around for quite a long time, but an actual smart home has only established a short time. The invention of home appliances such as a television with a remote control which is a simple home automation system was patented in 1893. Other than that, more home appliances have been invented since that. In early 2000, the popularity of smart home automation began to increase due to different of technology started to arise. Smart home automation turns into a more affordable choice and therefore a viable or available technology for consumers. With the uprising of the prestige of smart home, home networking, domestic technologies and other gadgets began to emerge on store shelves Today's smart home automation are more emphasize about smart living, living greener and security.

Our smart home is sustainable and it ensures that our home is not utilizing unnecessary energy. In addition, the smart home also can prevent any intruders

by alerting us with alarm or send any signal to us through smartphone related application. The current trend in smart home automation includes automated lights, remotely mobile control, remote video surveillance and receiving notifications of mobile, email and text.

2.3 Problem Statement

Nowadays people with the hectic daily life routine sometimes makes them forgetful to switch off the devices at home. As a human being we can't run from the clumsiness attitude plus with our packed daily routine life that sometimes makes ourself such in hurry situation that sometimes makes us forgot to switch off the lamps.

It will cause the electricity bill rose sharply. Besides it is one of the electricity wastage that will lead the earth to became an unhealthy ones. Besides, the elderly and the handicapped user faced problem to manually access control of light and fan instead of automation process.

2.4 Research Objective

The main objective of this project is to control the devices such as lamp and door at home using smartphone.

1. To design one system that is related with home appliances using NODEMCU

2. To implement a home appliances that can help user to control the devices at home.

3. To develop a good condition of house area that will prevent any loss and damage to the property of any organization.

2.5 Project Significance

Most of the project that is related with home automation or known as home appliances most of it have been done by using the blynk apps and rarely use the telegram to give the command. The differences in my project is NODEMCU is being used compare to others like they are using WIFI module only and most of it Is using Arduino, Bluetooth, ZigBee and GSM. The similarity is we are using IOT as a platform to control the devices at home which is our devices need to connected to internet first before wanted to give command through telegram or blynk application. As in my project I am using smartphone to give command compare to the other project that is using tablet laptop and others

Chapter-3: System Development

3.1 Tools used

1. **Wi-Fi module ESP8266 -:**

   The Wi-Fi module ESP8266 is a minimized effort Wi-Fi chip which has full TCP/IP stack & microcontroller capacity created by Espressif Systems which is Shanghai based Chinese maker. This chip went for the consideration by western producers in Aug 2014 along with the ESP-01 module, created by some outsider Ai-Thinker. This tiny chip helps microcontrollers to interact with a Wi-Fi system and make TCP/IP associations. Notwithstanding, during that time there no was no English documentation on the chip and the directions acknowledged by it. The specific lesser cost & the way there were not a lot of exterior parts on the module, which suggested that it could in the long run be reasonable in volume, used by numerous programmers to investigate the module, chip, and the product on it.

Figure 3.1.1 WIFI Module ESP8266

2. Relay Module –:

A relay circuit is used as a electromagnetic switch to turn on or off the electrical gadgets connected with 220v supply.

Input consists of :

VCC according to the tradition is associated with the positive supply voltage. Ground is set apart as GND which the name tells, is connected with -ve terminal of the supply voltage IN1 and IN2 are the correspondence pins Each sub-module in the transfer has one NC (typical close) & one NO (ordinary open).

So hand-off accompanies 2 NC, 2 NO and 2 COM of the direct hand-off altogether. NC is an abbreviation for the state without power and the typical close port contact. NO is an abbreviation for the state with power and the typical open port contact. COM represents the common port.

Figure 3.1.2 Relay Module

3. **Jumper wires -:**

Jumper wires are simply wires that have connector pins at each end, allowing them to be used to connect two points to each other without soldering. Jumper wires are typically used with breadboard and other prototyping tools in order to make it easy to change a circuit as needed. Fairly simple. In fact, it doesn't get much more basic than jumper wires.

Figure 3.1.3 A Few Jumper Wires

3.2 Software Used

1. **Arduino IDE -:**

The Arduino (IDE) is a cross-platform application for which java programming language is used in writing it. It is used for writing and uploading programs into the Arduino compatible boards in an iot projects. Python is used as coding language for the programs. The source code of the IDE is discharged under the GNU General Public License, rendition 2. The Arduino IDE bolsters the dialects C and C++ utilizing uncommon guidelines of code organizing. The Arduino IDE supplies a product library from the Wiring project, which gives numerous basic info and yield techniques.

Client composed code just requires two fundamental capacities, for beginning the sketch 27 and the principal program circle, that are arranged and connected with a program stub primary () into an executable cyclic official program with the GNU tool chain, additionally included with the IDE appropriation. The Arduino IDE utilizes the program avrdude to change over the executable code into a content document in

hexadecimal encoding that is stacked into the Arduino board by a loader program in the board's firmware.

Figure 3.2.1

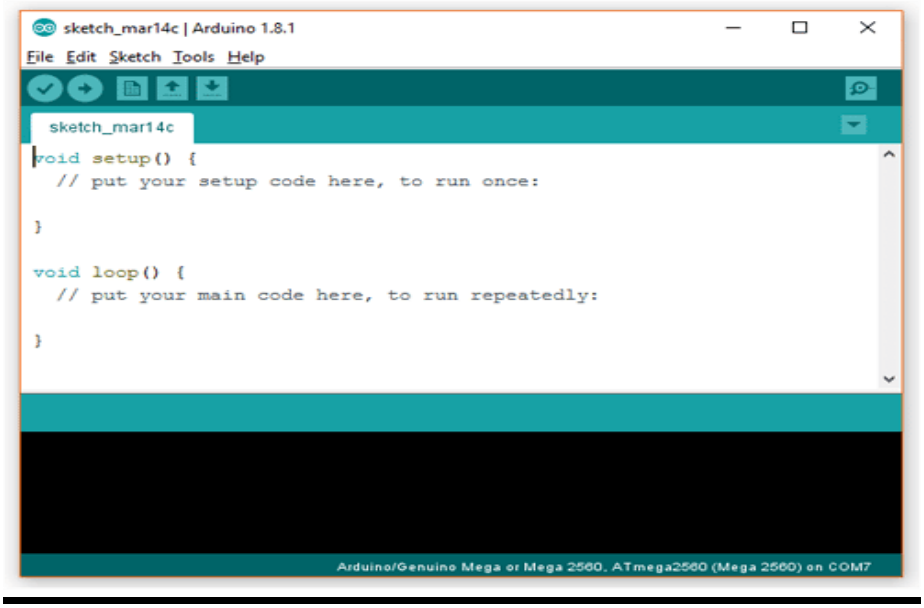

Figure 3.2.2

2. Cadio Home Automation-:

Figure 3.2.3

Chapter-4: Methodology

4.1 Schematic Diagram: -

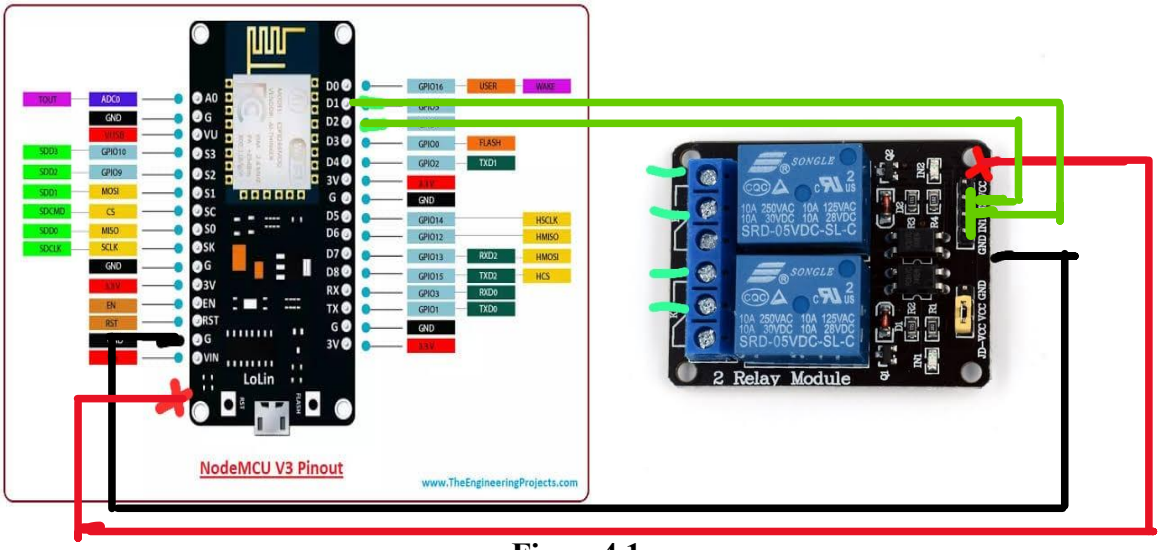

Figure4.1

4.2 Block Diagram

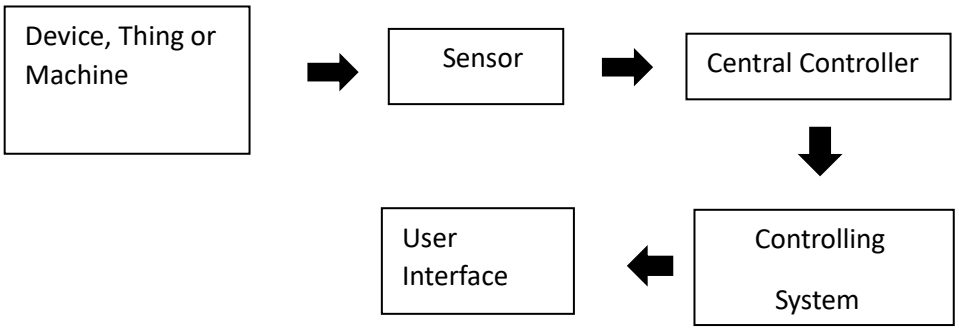

Figure 4.2

4.2 Algorithm Of the Program

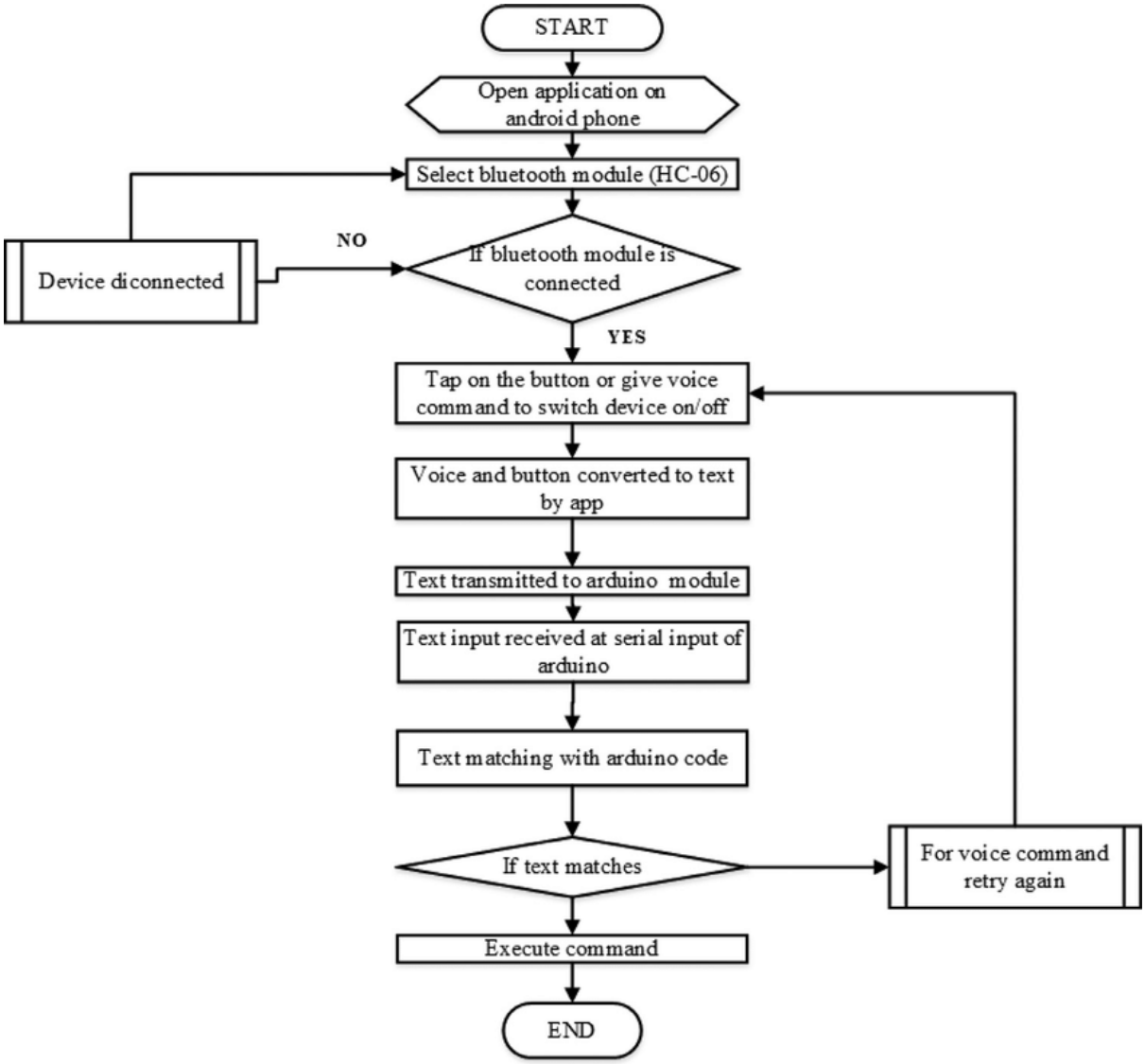

Figure 4.3

Chapter 5-: Result

1. **When the light is switched on**

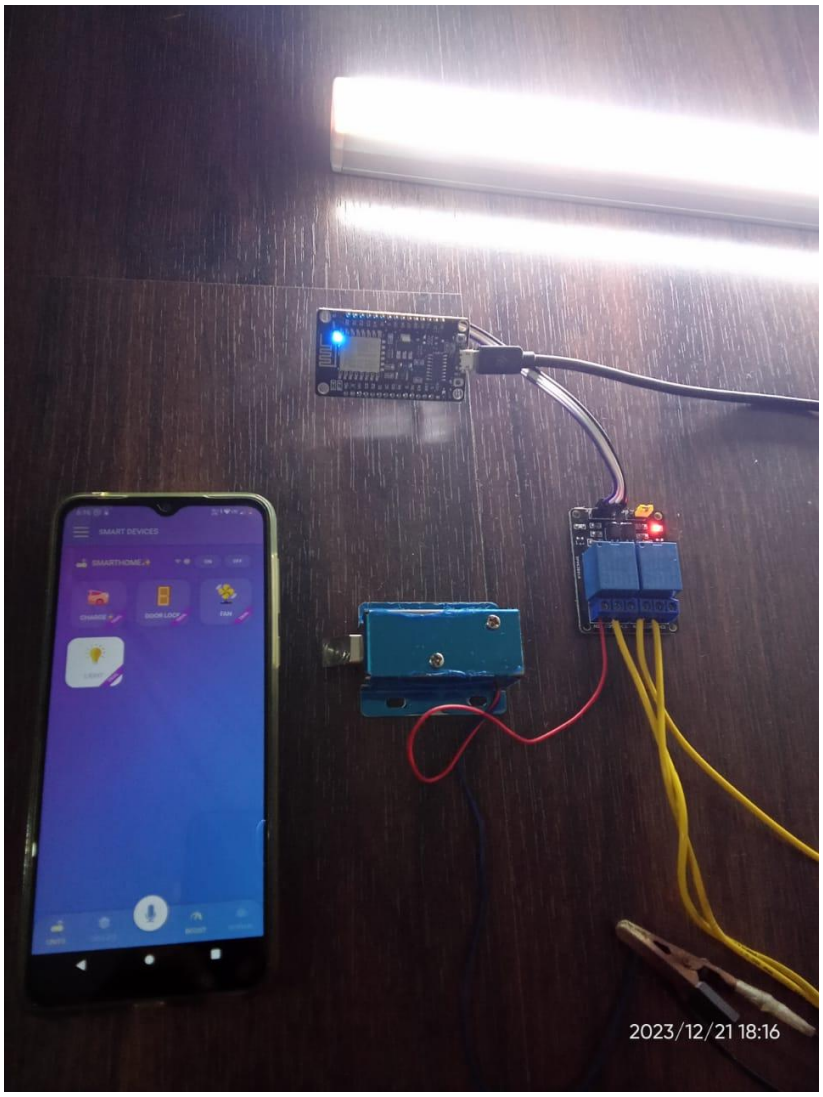

Figure 5.1

2. When the light is switched off

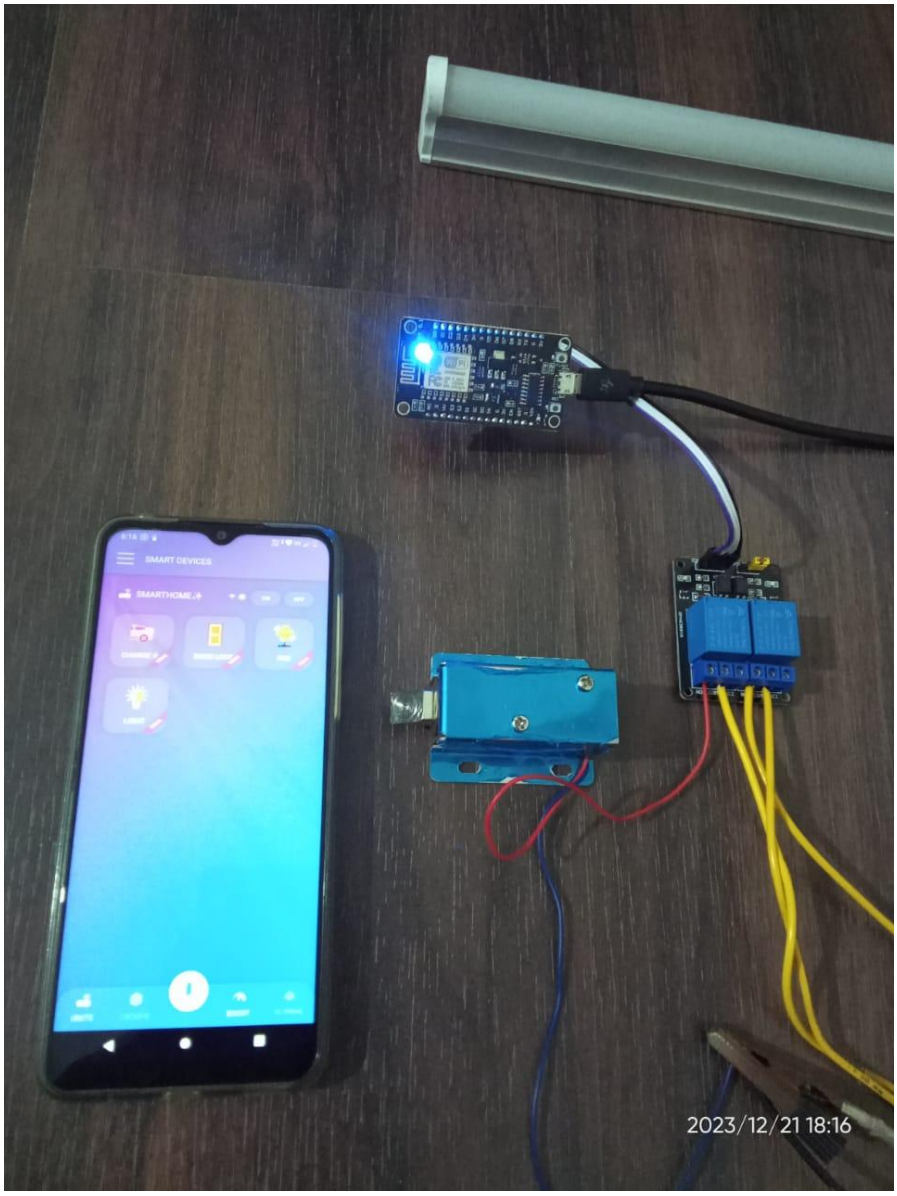

Figure 5.2

3. **When the door lock is off**

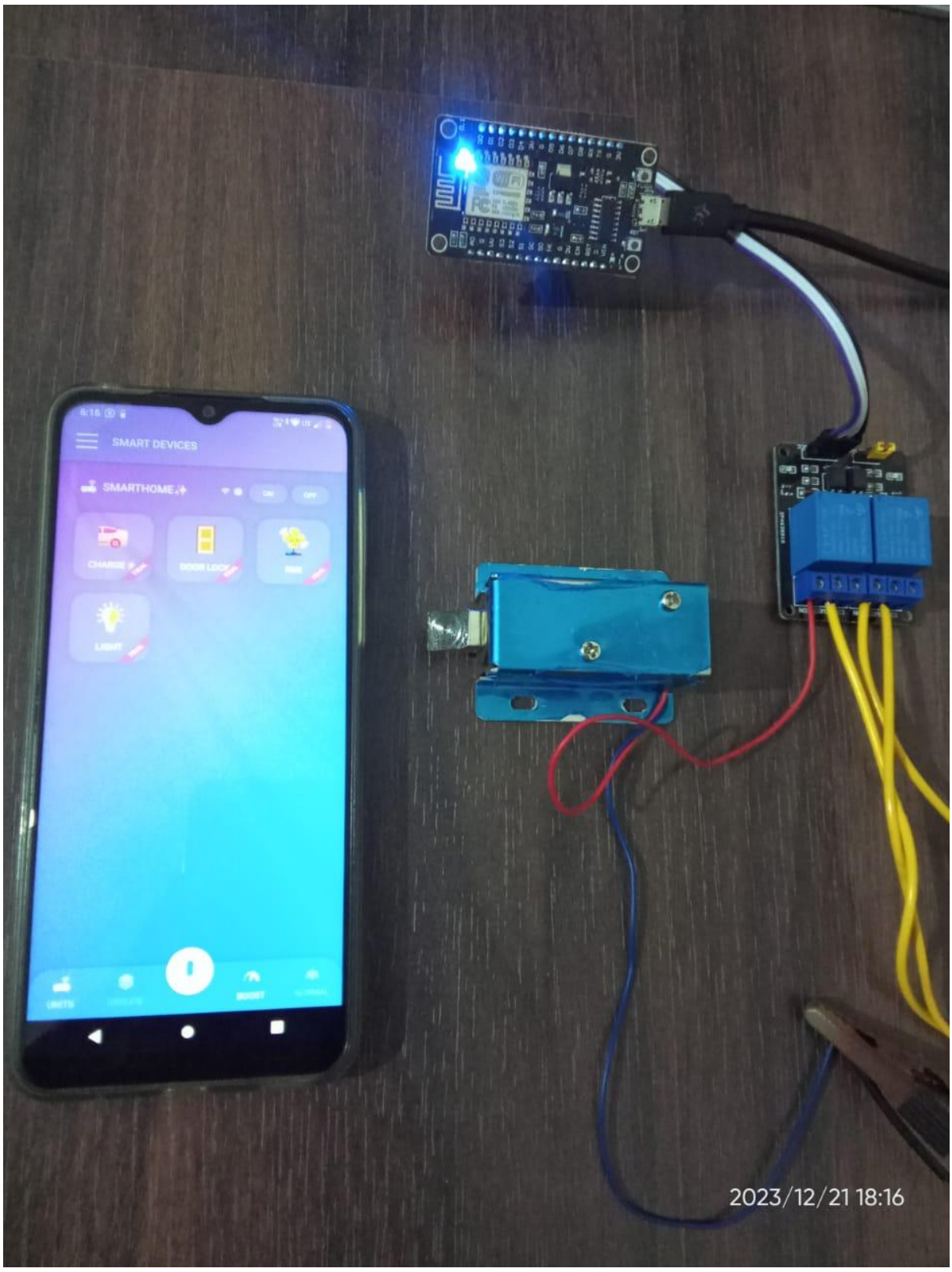

Figure 5.3

4. When the door lock is on

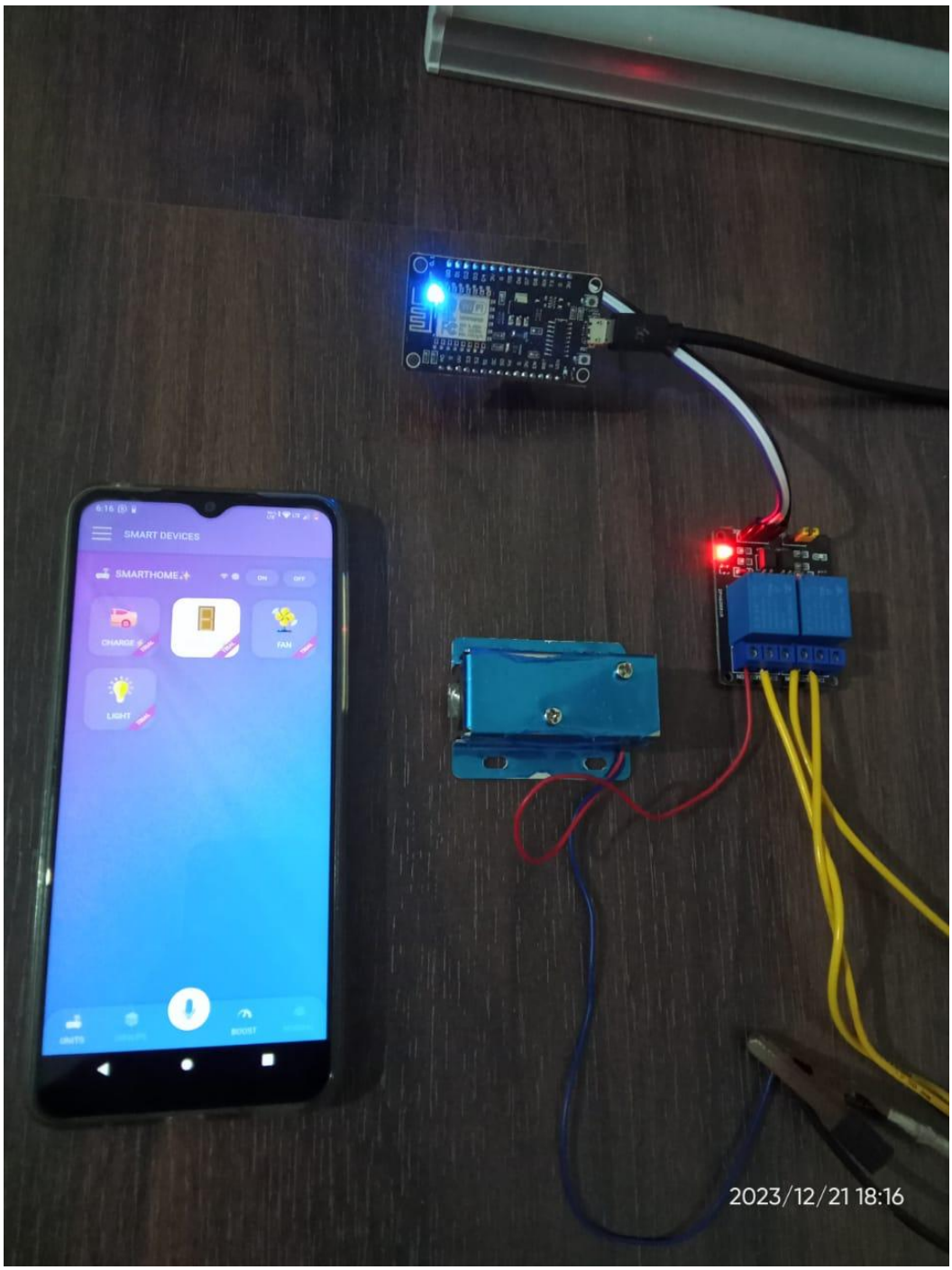

Figure 5.4

Chapter 6-: Advantages and Disadvantages

6.1 Advantages

Accumulating Convenience in everyday Life –

When we change our home into a shrewd house, we'll have every one of our products customized to our specific needs. Also having the capacity to deal with our home, regardless of where we are, it tends to be massively advantageous.

Customization –

There are a great deal of keen wares in the commercial center at present and we certainly don't require obtaining these without a moment's delay. As the buyer it is absolutely up to us to make a choice which item we need fundamentally, decide whether we like it, and afterward include it to our arrangement of savvy home products as we go. An astounding item in the first place may be an indoor regulator or it may be a house security framework in case we're in the commercial center for both of them.

Security –

Shrewd house security frameworks allow us to break down our home regardless of where we are. We could have introduced cameras, or introduced movement finders, and locks, and so forth, and we would be informed immediately on the off chance that anything is extraordinary or remarkable. A great deal of these frameworks would even enable us to distinguish us of any unforeseen temperature adjustments so we're alarmed in the event that there is a potential fire.

Simplicity of Utilization –

Generally all savvy house wares can be placed in excluding a lot of problem, a few of them don't require us to convey somebody into our home. Also in the event that we are as of now a man who is an innovation insightful, learning of how to use to a great extent every one of these products is a puff of air.

Saving Money & saving Environment –

Savvy house highlights wares like climate control systems .indoor regulators, and lighting. By having the capacity to set this stuff on premise of a clock or to turn it on or off at whatever point we're far from our home will most likely help

us spare our cash on our electric bills. A considerable measure of these items would enable us to trail our vitality tradition and uses.

6.2 Disadvantages

Cost –

The majority of the families nowadays are skilled to purchase keen house products, yet it doesn't imply it would not put down a gigantic imprint in your financial plan. You can likewise purchase the items one by one and after that it would not appear its excessively, $300 there, $50 here, however nearly by the occasion when you contain the shrewd house association we need, we would most likely have depleted a bigger sum than we would have in the event that we had acquired non-savvy items.

Slight Learning Curve –

We realize we settled on the focal points like the majority of the brilliant home frameworks being in reality exceptionally easy to utilize, anyway at the comparative time there exists to some degree a learnedness bend for the vast majority of the general population. For any individual who is now engaged in innovation, changing your savvy house would be a puff of air, yet for other people, who are not all that tech nous, it may lead for a swarm of time which would be spent perusing manuals.

In the event that we figure we may have issues about figuring out how would we utilize shrewd house gadget, the answer might be direct. Request help! Addressing somebody to show it to us how to rush your brilliant house have the capacity to remotely less confounding than attempting to make tails or leaders of an exercises manual.

Reliability –

A shrewd house will be tremendously subject to our web association. On the off chance that our affiliation or connection drops we'll be run or left with an arrangement of keen items that won't work. Moreover, remote signs can most likely be separated by previous gadgets in our home and it would make a portion of our brilliant items work with a moderate fast or not in the least.

There are heaps of focal points and impediments which we need to consider while choosing whether we'd need to change our home into a keen house. Savvy houses are not for everybody. They can help cutting down our power installments, compress our day by day assignments, and helped to give ourselves a sentiment of security. In any case, for various individuals keen house produced products will only change into a money related weight. It is in the long run up to you to pick which space you fall into

Chapter 7-: Future Scope

Utilizing this framework as structure, the framework can be extended to incorporate different alternatives which could incorporate home security include like catching the photograph of a man entering and sending it to the proprietor through WhatsApp or executing the entryway bolt framework. This will expand the security and empower to keep a mind his home from wherever. There will be sensors that will take a shot at constant to open the entryway when an approved individual is inside stipulated range from the entryway. The framework can be extended for robotization of different gadgets at home.

SMART HOME EXPANSIONS

o **Wireless connectivity:** A center point which is wifi associated that isn't having any physical association with switch can be a decent improvement, as this gives you greater adaptability to pick a suitable area for it in the house. o Scalability: Hub should be supporting a big number of devices for achieving full automation.

Figure 7.1

o **Test for compatibility:** A center point that we utilize must have the capacity to help & offer similarity to the real conventions like Bluetooth; zwave and so forth. Furthermore, similarity with the gadgets around the house ought to be tried.

o **App control:** Web interfaces look appealing yet portable applications are a lot simpler to work from the perspective of a client so ensuring application perfect with 40 every advanced mobile phone is created.

o **Scheduling the devices and actions:** The framework must have the capacity to draw plans concerning different gadgets and make activities to associate these hubs.

o **Real time Alerts/Messaging:** The planned framework ought to have the capacity to transmit the alarms when client activities are practiced, for example an alarm message when somebody opens fundamental entryway of the house with her verified key

Chapter 8 -: Conclusion

Nowadays, technologies is growing rapidly and it is same goes to the home automation. As for this project it highly recommended for everyone in this world especially for a user who is with disabilities and for the householder too. This recommendation will lead to a green world which is it can help to save and reduce on electricity bills. Plus, it will help and lead the disabilities person can work independently and help them to manage their house safety in more organized way. It will help to prevent any loss to a user or unwanted circumstances to a user

The house mechanization by methods for Internet of Things has now been tentatively checked to exertion pleasantly by connecting uncomplicated apparatuses to the framework and the machines were viably controlled through the web. The expected framework not simply checks every one of the information of sensors, similar to gas, temperature, movement sensors, light however it additionally enacts a technique in understanding to the necessities, for example, exchanging the lights on when it turns dull. It besides supplies the sensor parameters to the made database in an all around planned way. This will help the shopper to investigate the request of an assortment of strictures at home whenever and anyplace.

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